Math 141, F11 | Name: Answer
Quiz 10 Section: | SSN: xxx-xx-

<u>Instructions:</u> There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz and exam.

**Problem 1.**(10 points) Find two positive numbers whose product is 81 and whose sum is a minimum.

Solution: $S = x + y, \quad x > 0, y > 0$

$x \cdot y = 81$ gives us $y = \frac{81}{x}$.

then we get

$$S = x + \frac{81}{x}, \quad x > 0$$

$$S'(x) = 1 - \frac{81}{x^2}$$

$$S'(x) = 0 \Rightarrow 1 - \frac{81}{x^2} = 0$$

$$\frac{81}{x^2} = 1$$

$$x^2 = 81$$

$$x = 9 \quad \text{Since } x > 0.$$

It is obvious $S'(x) < 0$ when $0 < x < 9$

$S'(x) > 0$ when $x > 9$.

Thus $S(9) = 9 + 9 = 18$ is the absolute minimum

in which $x = 9$, $y = 9$.